Absolute Error and MAE

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable loss-and-unit evidence record.

Trace row errors,compute distinct averages and convert target and squared units correctly.

Keep one habit: W1–W3/W5 → selected metre task → numeric project loss record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Mean, Median, and Baselines

> **My remembered idea, example and correction**

Error Analysis

> **My remembered idea, example and correction**

Confusion Matrix

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main residuals [-1,2,-3] cm and pictured [-1,2,3] cm have absolute magnitudes [1,2,3] and squares [1,4,9]: MAE 2 cm,MSE 14/3 cm²,RMSE sqrt(14/3)≈2.16 cm. cm→mm gives MAE20 mm,MSE1400/3 mm²,RMSE≈21.60 mm. Lab training [5,7,9] cm fits mean7; E1–E3 observed [10,10,10],candidate [10,10,4],baseline [7,7,7]. Candidate MAE2/MSE12/RMSEsqrt12; baseline MAE3/MSE9/RMSE3,with target/squared units explicit.

- Reference for this exercise: open this week's student lesson, "Absolute Error and MAE", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Calculate MAE for the worked residuals.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main residuals [-1,2,-3] cm and pictured [-1,2,3] cm have absolute magnitudes [1,2,3] and squares [1,4,9]: MAE 2 cm,MSE 14/3 cm²,RMSE sqrt(14/3)≈2.16 cm. cm→mm gives MAE20 mm,MSE1400/3 mm²,RMSE≈21.60 mm. Lab training [5,7,9] cm fits mean7; E1–E3 observed [10,10,10],candidate [10,10,4],baseline [7,7,7]. Candidate MAE2/MSE12/RMSEsqrt12; baseline MAE3/MSE9/RMSE3,with target/squared units explicit.

- Give MSE and **RMSE** with **unit**s.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main residuals [-1,2,-3] cm and pictured [-1,2,3] cm have absolute magnitudes [1,2,3] and squares [1,4,9]: MAE 2 cm,MSE 14/3 cm²,RMSE sqrt(14/3)≈2.16 cm. cm→mm gives MAE20 mm,MSE1400/3 mm²,RMSE≈21.60 mm. Lab training [5,7,9] cm fits mean7; E1–E3 observed [10,10,10],candidate [10,10,4],baseline [7,7,7]. Candidate MAE2/MSE12/RMSEsqrt12; baseline MAE3/MSE9/RMSE3,with target/squared units explicit.

- Why do [0, 0, 6] and [2, 2, 2] have equal MAE but unequal MSE?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main residuals [-1,2,-3] cm and pictured [-1,2,3] cm have absolute magnitudes [1,2,3] and squares [1,4,9]: MAE 2 cm,MSE 14/3 cm²,RMSE sqrt(14/3)≈2.16 cm. cm→mm gives MAE20 mm,MSE1400/3 mm²,RMSE≈21.60 mm. Lab training [5,7,9] cm fits mean7; E1–E3 observed [10,10,10],candidate [10,10,4],baseline [7,7,7]. Candidate MAE2/MSE12/RMSEsqrt12; baseline MAE3/MSE9/RMSE3,with target/squared units explicit.

- A team switches from MSE to MAE only after seeing a bad large error. What should be reported?

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This independent task starts with residuals in metres. Apply each operation to the supplied three residuals before converting the final losses: metre-to-centimetre scale100 for MAE/RMSE and10000 for MSE. The independent values differ from the centimetre worked example. State squared units for MSE and do not infer new model predictions or unseen-case reliability.

- Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Compute squared **loss** for Absolute and squared **loss**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Absolute and squared **loss**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Actual=[3,5,9], predicted=[2,7,6] cm.
- Calculate MAE.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Actual heights are 4, 8 and 10 cm; **prediction**s are 5, 6 and 13 cm. Calculate **absolute error**s and MAE.
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 21 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Compare absolute and squared error without losing track of **unit**s.
- Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/21

Absolute Error and MAE

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Absolute errors

Absolute values1, 2, 3 give MAE6/3=2 cm.

2. Squared errors

Squares1, 4, 9 give MSE14/3 cm2, approximately = 4.67 cm2.

3. Root squared error

RMSE=sqrt(14/3), approximately2.16 cm.

CHANGE ONE THING

Convert every residual to millimetres.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Compare absolute and squared error without losing track of **unit**s.
- Which labelled step matches this detail? **RMSE**=sqrt(14/3), approximately2.16 cm.
- 1. **Absolute error**s
- 2. Root squared error
- 3. Squared errors
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Root squared error **RMSE**=sqrt(14/3), approximately2.16 cm. Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Convert every residual to millimetres. Which conclusion is supported by the supplied facts?
- 1. MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

<details>
<summary>Check the changed explanation</summary>

After your attempt, compare with the supported explanation: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm. Residuals scale by10; squares scale by100, while the square root returns a length **unit**.
</details>

Repair and keep the record

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Absolute Error and MAE: the case

AI ACADEMY / AI-4/21 · 01 · LOOK AT THE FACTS

Absolute Error and MAE: the case

The situation

Compare absolute and squared error without losing track of units.

Supplied evidence

Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain absolute error and mae using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

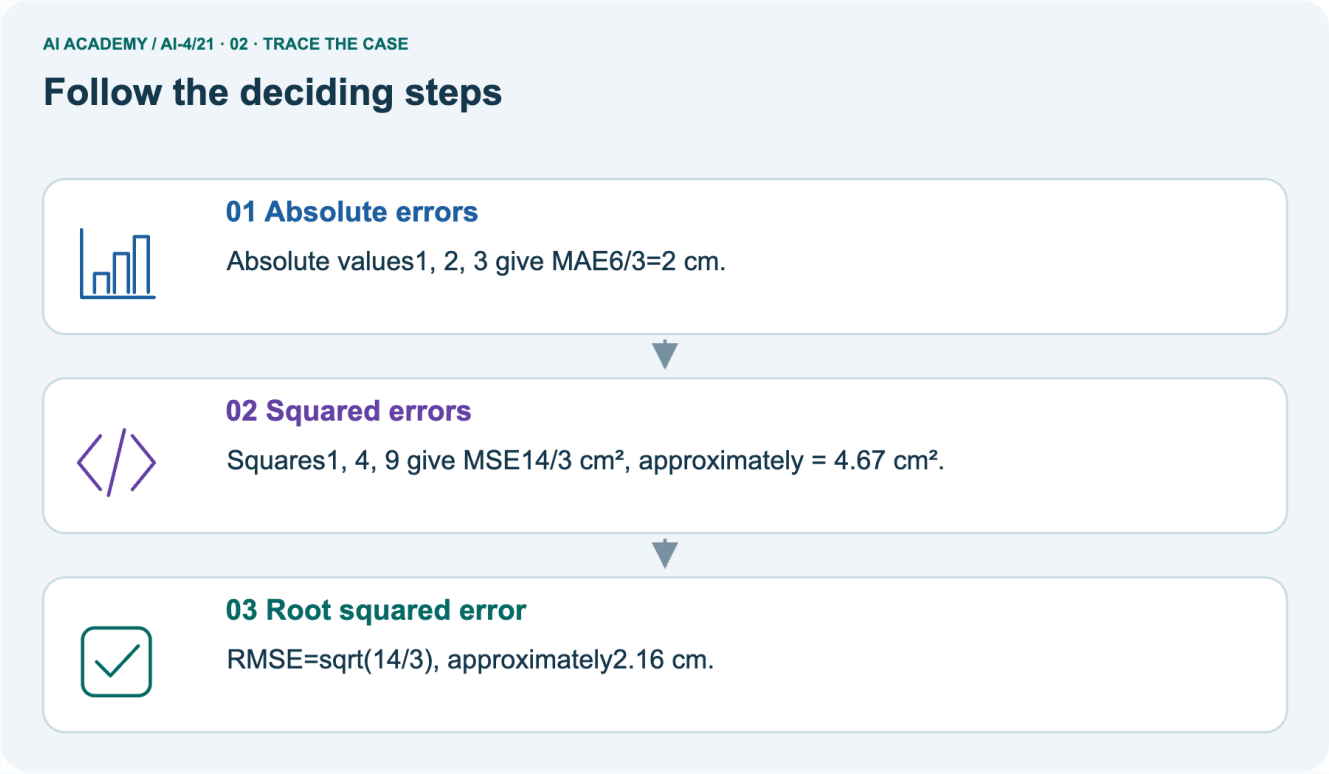
Compare absolute and squared error without losing track of **unit**s. Supplied facts: Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute errors: Absolute values1, 2, 3 give MAE6/3=2 cm. → Squared errors: Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm². → Root squared error: **RMSE**=sqrt(14/3), approximately2.16 cm.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Absolute and squared loss at a glance

AI ACADEMY / AI-4/21 · 03 · INSPECT THE LESSON MODEL

Absolute and squared loss at a glance

Part	Evidence or action	Interpretation
Residuals	minus 1 plus 2 minus 3	Signed differences
Absolute values	1 2 3	MAE 2 cm
Squares	1 4 9	MSE 14 divided by 3

Model A has residuals [0, 0, 6]; model B has [2, 2, 2]. Both MAEs are 2, but MSEs are 12 and 4. Squaring distinguishes concentrated large errors from evenly spread smaller errors. Which pattern matters depends on the task and consequences.

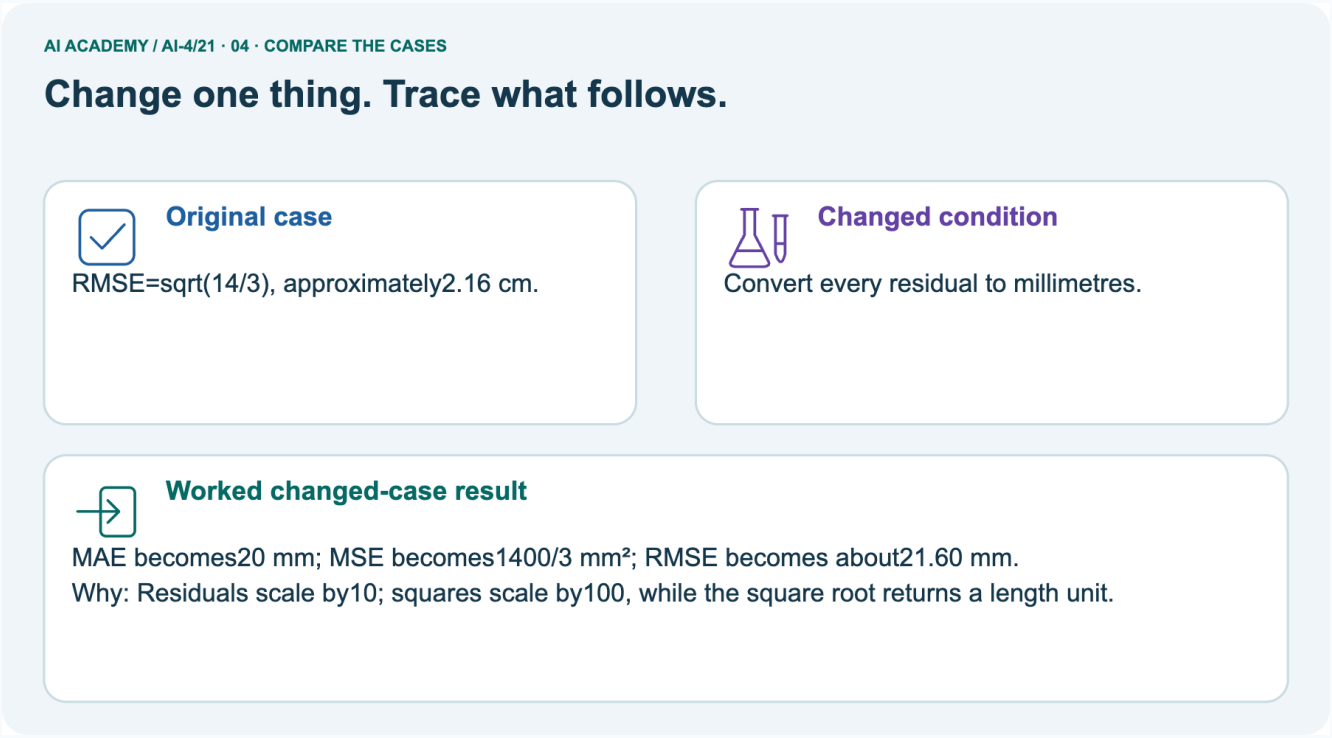
Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute and squared **loss** at a glance. Residuals | minus 1 plus 2 minus 3 | Signed differences. Absolute values | 1 2 3 | MAE 2 cm. Squares | 1 4 9 | MSE 14 divided by 3

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **RMSE**=sqrt(14/3), approximately2.16 cm. Changed condition: Convert every residual to millimetres. Expected result: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-4/21 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

Read the picture in order. RMSE=sqrt(14/3), approximately2.16 cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What it does not establish

Different loss functions weight large misses differently; a smaller selected loss is not automatically better for every consequence.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

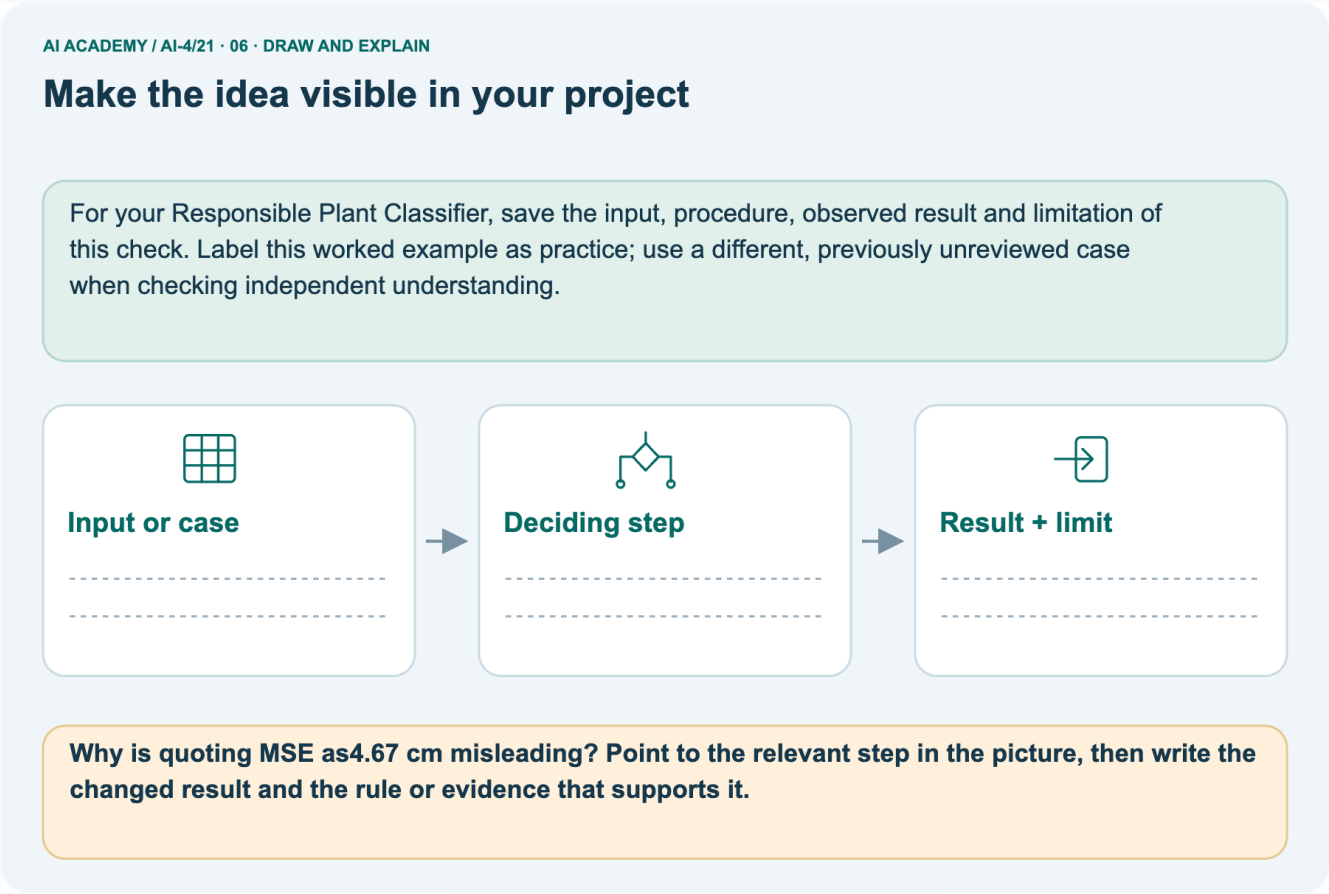
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Absolute Error and MAE: the case

02. Follow the deciding steps

03. Absolute and squared loss at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.